Austin Renz

Momentum vs. Mean Reversion across Sector ETFs, Horizons, and Volatility Regimes

Quant Mentorship Final Project: Summary Analysis Report

1. INTRODUCTION & QUESTION

This project asks a single, focused research question: how do momentum and mean-reversion behaviors in US equity ETFs vary across (i) the ETF itself, (ii) the return horizon, and (iii) the prevailing volatility regime? The aim is not to build a trading strategy, but to characterize where and when the textbook patterns of short-horizon reversal and longer-horizon persistence actually show up in real data.

2. DATASET & SOURCE

Daily adjusted close prices for six ETFs (SPY, QQQ, XLK, XLE, XLF, XLU), downloaded from Yahoo Finance (yfinance, auto_adjust=True) over 2010-01-05 to 2026-04-17 (4,248 trading days after cleaning). Adjusted closes are used so that splits and dividends are baked into the price series, making log-returns a clean measure of total return. The ETFs were chosen to combine a broad-market anchor (SPY), a growth anchor (QQQ), and four sector views of different parts of the economy (technology, energy, financials, utilities).

3. METHODS

- Cleaning: reindex to a business-day calendar, forward-fill gaps of at most one day, drop any remaining missing rows.
- Feature engineering: for each ETF and each horizon h in {1, 5, 20} trading days, construct two aligned columns:
  - ret_lag_h = cumulative log-return over the previous h days (observed at t),
  - ret_fwd_h = cumulative log-return over the next h days (realized strictly after t).

  This alignment is essential; it is what prevents lookahead bias.
- Volatility: 20-day rolling realized volatility, annualized.
- Regime indicator: the top/bottom quartile of SPY's 20d realized vol using expanding-window thresholds (so classification at date t depends only on history up to t).
- Statistical tests: Pearson and Spearman correlations between ret_lag_h and ret_fwd_h, full-sample and within each regime.
- Modeling: chronological 80/20 train/test split; a simple univariate OLS (fwd = a + b * lag) and an extended OLS that adds rolling vol, a high-vol dummy, and a lag x high-vol interaction; Newey-West (HAC) standard errors with 10 lags to correct for the overlap induced by the rolling forward-return target.

Austin Renz

Momentum vs. Mean Reversion across Sector ETFs, Horizons, and Volatility Regimes

Quant Mentorship Final Project: Summary Analysis Report

(continued)

4. KEY FINDINGS

(a) SHORT-HORIZON MEAN REVERSION IS PERVASIVE. Over the full sample, the Pearson correlation between past 1-day and next 1-day returns is negative for every ETF, ranging from -0.022 (XLE) to -0.105 (XLK) and -0.100 (XLF). The 1d reversal is most pronounced in broad-market and sector ETFs that are heavily traded.

(b) 20-DAY RETURNS ALSO MEAN-REVERT, ESPECIALLY XLU. At the 20-day horizon, five of six ETFs show correlations between -0.07 and -0.22. Utilities (XLU) is the strongest: r = -0.223, meaning streaks of outsized 20-day moves in XLU are followed on average by partial give-back. Broad-market SPY sits at r = -0.146.

(c) XLE IS THE NOTABLE OUTLIER. Energy is the only ETF with a positive 5-day correlation (r = +0.026), and its short-horizon reversal is the mildest of the group. This is consistent with XLE's heavy exposure to commodity prices, which exhibit their own momentum dynamics.

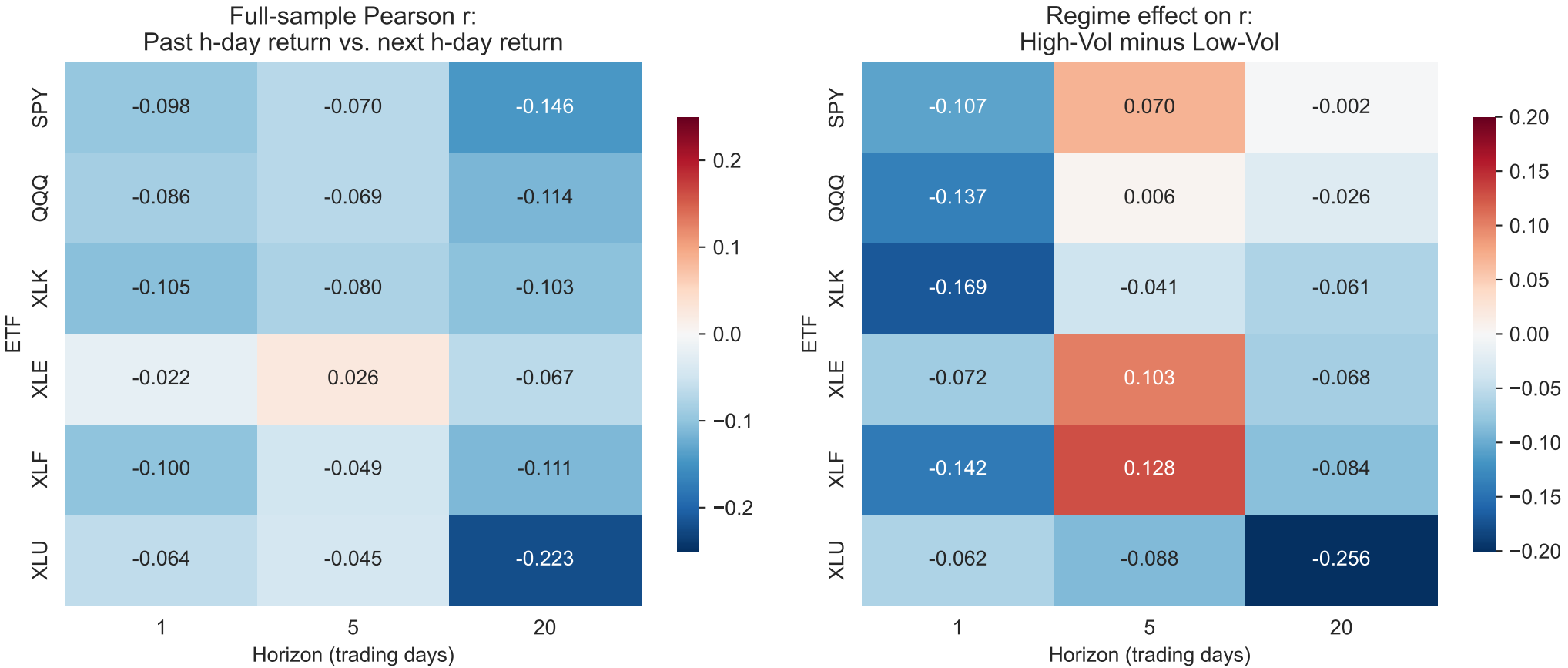
(d) HIGH VOLATILITY DRAMATICALLY AMPLIFIES SHORT-HORIZON REVERSAL. At h = 1 day, the Low-Vol regime shows nearly flat correlations (-0.042 to +0.024), while the High-Vol regime shows sharp negatives: SPY -0.149, QQQ -0.149, XLK -0.174, XLF -0.160, XLU -0.104. This is the single cleanest result in the study.

(e) REGRESSIONS CORROBORATE THE STORY. The univariate lag-return slope is negative for every sector at the 5-day horizon, and the HAC-adjusted t-statistics on the extended SPY regression are consistent with the sign pattern, though the R^2 is small (as expected for daily equity returns).

5. REFLECTION & LIMITATIONS

The effect sizes are statistically visible but economically small on a per-trade basis; real strategies combine many such weak signals and have to survive costs and frictions not modeled here. The regime split depends on the chosen volatility window and quartile cutoffs; other choices would shift magnitudes without overturning the sign pattern. Overlapping 5- and 20-day forward-return windows induce residual autocorrelation; we address it with HAC standard errors but do not eliminate it. Most importantly, the panel ends in 2026 and covers one extended low-rate era (2010-2021) and one tightening era (2022-2026); different economic regimes could produce different numbers. The central takeaway is that mean reversion's strength depends jointly on horizon, sector, and volatility regime. That result is robust to these caveats even if the exact coefficients are not.

Figure 1. Horizon and regime dependent predictability of sector ETF returns (2010-2026)



Left: negative values indicate short-horizon mean reversion. Right: blue cells show that predictive sign strengthens (toward reversal) in high-vol regimes.

Figure 2a. Growth of $1 invested, 2010-2026 (adjusted close)

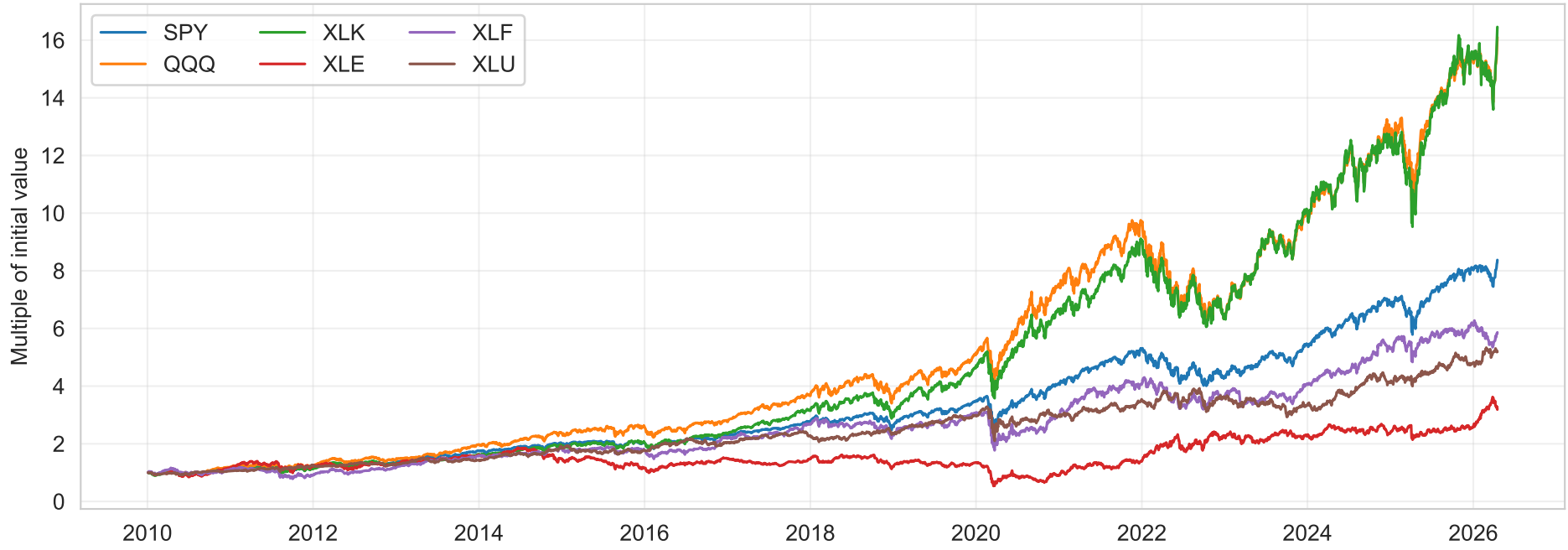
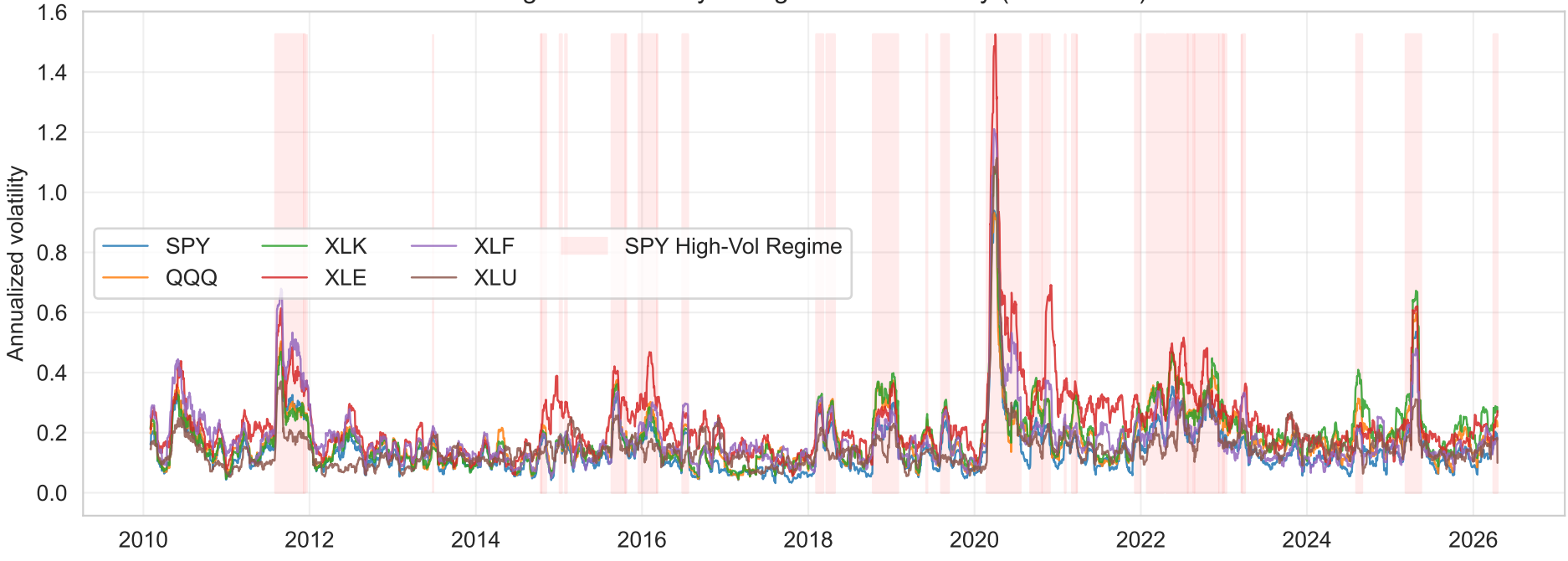



Figure 2b. 20-day rolling realized volatility (annualized)